GLM2 Explainer: Logistic Regression, Log-Odds, and the Interaction Model

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1. Why not just use percent correct directly?

The outcome in each trial is binary: either the observer correctly identified the translation direction (1) or did not (0). Percent correct is just the mean of these 0/1 values across trials. It is intuitive and we report it throughout.

The problem with fitting a standard linear regression to a binary outcome is that it can predict probabilities below 0% or above 100%, and the equal-variance assumption is badly violated near floor and ceiling. More fundamentally, the relationship between a predictor and the probability of success is inherently nonlinear: a 10pp increase from 12.5% (chance) to 22.5% reflects a much larger change in underlying signal than a 10pp increase from 45% to 55%.

Logistic regression solves this by modelling a transformation of the probability — the **log-odds** — rather than the probability itself. On the log-odds scale, the model is linear, well-behaved at all probability levels, and gives directly interpretable coefficients.

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2. From probability to odds to log-odds

Probability (p): familiar — ranges 0 to 1.

Odds: the ratio of success to failure — p / (1 – p).

- p = 0.125 (chance, 1/8 correct) → odds = 0.125/0.875 = **0.143**
- p = 0.25 (25% correct) → odds = 0.25/0.75 = **0.333**
- p = 0.50 (50% correct) → odds = 0.50/0.50 = **1.000**
- p = 0.75 (75% correct) → odds = 0.75/0.25 = **3.000**

Odds range from 0 to ∞. A doubling of the odds means the same proportional improvement in performance regardless of where on the scale you start — it is a multiplicative, not additive, measure.

Log-odds (also called the *logit*): the natural logarithm of the odds — log(p / (1 – p)).

- p = 0.125 → log-odds = log(0.143) = –**1.946**
- p = 0.25 → log-odds = log(0.333) = –**1.099**
- p = 0.50 → log-odds = log(1.000) = **0.000**
- p = 0.75 → log-odds = log(3.000) = **+1.099**

Log-odds range from –∞ to +∞, with zero at 50% correct. Negative values mean below 50%; positive values mean above 50%. The logit is symmetric: p=0.25 and p=0.75 are equidistant from 50% on the log-odds scale (±1.099), even though 25% is closer to floor and 75% is closer to ceiling.

Converting back: given a log-odds L, the probability is p = 1 / (1 + e^(–L)).

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3. The logistic regression model

Logistic regression assumes the log-odds of a correct response is a linear combination of the predictors:

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log-odds(correct) = beta0 + beta1·F1 + beta2·F2 + beta3·F3 + beta4·F4
                       + beta5·(F1xF2) + beta6·(F1xF4) + beta7·(F2xF4)
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The *beta* coefficients are estimated by maximum likelihood — they are the values that make the observed data most probable under the model. This is the linear model the user expected: one coefficient per term, estimated directly from the data.

The four predictors are all binary (0/1):

Factor	= 1 when	= 0 when
F1 Dot cueing	Trial is CUED (temporal onset cue marks translator)	UNCUED
F2 Depth-field cued	Translator depth plane matches delayed field's onset depth	Mismatch
F3 Color-field cued	Translator color matches delayed field's original color	Mismatch
F4 Translator Near	Translator occupies the Near depth plane	Far

F2=1 conditions: CUED+N, CUED+C, UNCUED+Z, UNCUED+CZ.

F2=0 conditions: CUED+Z, CUED+CZ, UNCUED+N, UNCUED+C.

(F3 follows the complementary pattern; F4 depends on b_near and cond.)

Three interaction terms are included: F1×F2 (dot × depth), F1×F4 (dot × translator depth), F2×F4 (depth × translator depth). These test whether the effect of one factor depends on the level of another.

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4. The estimated coefficients (n = 2051, all 4 sessions)

Term	*beta* (log-odds)	SE	z	p	Odds ratio
Intercept (*beta*0)	–0.713	0.137	–5.22	<.001 *	0.49
F1 Dot cueing	–0.287	0.174	–1.65	.098 n.s.	0.75
F2 Depth-field cued	–0.332	0.174	–1.91	.057 n.s.	0.72
F3 Color-field cued	+0.049	0.103	+0.47	.639 n.s.	1.05
F4 Translator Near	–0.833	0.185	–4.51	<.001 *	0.43
F1 × F2	**+1.785**	0.215	+8.30	<.001 *	**5.96**
F1 × F4	+0.488	0.221	+2.21	.027 *	1.63
F2 × F4	–0.675	0.216	–3.12	.002 **	0.51

McFadden pseudo-R^2 = 0.092. LRT $chi^2(7)$ = 228, $p < 10^{-45}$.

Reading the intercept

*beta*0 = –0.713 is the log-odds of a correct response when **all predictors = 0**: UNCUED, depth anti-cued, color anti-cued, Far plane. Converting: p = 1/(1 + e^0.713) = **0.328**, or about 33%. This is the model's predicted accuracy for the "most anti-cued" Far-plane condition. It is not the overall baseline; it is the specific reference cell.

Reading a main effect coefficient

*beta*1 = –0.287 for F1 means: **when F2 = 0, F4 = 0, and all other predictors are at their reference level**, being CUED (F1 = 1 vs 0) changes the log-odds by –0.287. This is a small, non-significant, slightly negative number — surprising if you expected dot cueing to always help.

The reason it is near zero is that this coefficient captures the effect of dot cueing *only in the depth-anti-cued, Far-plane context* (the F2=0, F4=0 reference cell). In that specific context — where the depth cue works against the dot cue — being dot-cued provides little benefit. The large benefit of dot cueing emerges only when depth cueing is also present, which is captured by the interaction term.

Reading an interaction coefficient

*beta*5 = +1.785 for F1×F2 means: **the effect of F1 increases by 1.785 log-odds units when F2 = 1**. In other words, being dot-cued is 1.785 log-odds units more effective when depth is also cued than when it is not.

To find the total effect of being simultaneously dot-cued AND depth-cued (vs UNCUED and depth anti-cued), sum the relevant coefficients:

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Total effect of (F1=1, F2=1) vs (F1=0, F2=0):
  F1 + F2 + F1xF2 = –0.287 + (–0.332) + 1.785 = +1.166 log-odds
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Converting the reference cell (F1=0, F2=0, F4=0): intercept –0.713 → p ≈ 33%.

Adding 1.166 log-odds: –0.713 + 1.166 = +0.453 → p = 1/(1+e^–0.453) ≈ **61%**.

That shift from 33% to 61% is the full dot+depth cueing benefit, expressed in actual predicted accuracy. It matches what we observe in the raw data for CUED+N and CUED+C.

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5. Why the main effects look negative — the reference cell issue

In a model without interaction terms (GLM1), F1 = +22pp because that coefficient averages over all depth-cueing contexts. In GLM2 with an F1×F2 interaction term, the main effect of F1 is redefined as the effect *at F2 = 0*. Because the reference level (F2 = 0) is the anti-cued context, F1's main effect in GLM2 is the dot-cueing benefit in the worst-case depth scenario — which is near zero.

This is not a contradiction. Both estimates are correct; they answer different questions:

- GLM1 F1 = +22pp: *On average across all depth-cueing contexts, how much does dot cueing help?*
- GLM2 F1 = –5pp (AME): *Averaging the marginal effect of F1 over all observations (which includes the steep interaction slope), what is the average contribution of F1?* This is now small because most of F1's benefit is attributed to the F1×F2 term.
- GLM2 *beta*1 = –0.287: *At F2 = 0, F4 = 0, how much does F1 help?* This is the conditional coefficient in the specific reference cell.

The additive model (GLM1) was not wrong — it correctly identified which factors matter. But it attributed the joint F1∧F2 benefit to two separate main effects rather than a single synergistic interaction. GLM2 shows that the signal structure is: **dot cueing and**

depth-field cueing conjointly drive performance, and neither is effective in isolation.

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6. Average marginal effects (AMEs)

Because the logistic function is nonlinear, the same coefficient produces different probability-scale effects depending on where you are on the curve. Near $p = 0.5$ (the steepest part), a 1-unit change in log-odds produces a large probability change (~25pp). Near the floor ($p \approx 0.125$), the same 1-unit change produces a smaller probability change.

The **average marginal effect (AME)** for a predictor is the average, over all $n = 2051$ observations, of the derivative of P(correct) with respect to that predictor. For binary predictors, it approximates the average change in predicted probability when that predictor flips from 0 to 1, holding all other predictors at their observed values. The AME is expressed in percentage points.

Term	AME (pp)	interpretation
F1 Dot cueing	–5.3 pp	small, n.s., captured by interaction
F2 Depth-field cued	–6.1 pp	small, n.s., captured by interaction
F3 Color-field cued	+0.9 pp	null
F4 Translator Near	–**15.3 pp**	Near plane reliably worse
F1 × F2	**+32.7 pp**	synergistic conjunction
F1 × F4	**+8.9 pp**	dot cueing partly offsets Near penalty
F2 × F4	–**12.4 pp**	depth cueing less effective when Near

The AME for F1×F2 = +32.7pp means: when both F1 and F2 flip from 0 to 1 simultaneously (i.e., going from UNCUED+depth-anti-cued to CUED+depth-cued), the average predicted probability of a correct response increases by about 33 percentage points. This is the dominant effect in the model.

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7. Summary in plain language

The model asks: can we predict trial-by-trial accuracy from four binary features of the trial (dot cued?, depth cued?, color cued?, Near plane?), including three two-way interactions?

What we found:

1. **Dot cueing and depth-field cueing interact synergistically** (+32.7pp, the largest effect). Performance is high only when *both* the dot temporal cue and the depth-plane cue point to the same field. Neither alone does much. This is equivalent to saying: depth swaps (Z, CZ) disrupt dot cueing specifically because they sever depth-plane continuity of the attentional object.
2. **Color is completely null** (AME = +0.9pp, $p = .64$) across every model specification. The field's color identity provides no information for this task.
3. **Near-plane translation is intrinsically harder** (–15.3pp, $p < .001$), regardless of cueing. The Far > Near asymmetry is a robust, depth-specific effect (absent in monocular sessions from DepthSwapCtrl).
4. **The additive model (GLM1) was structurally misleading.** It estimated F1 = +22pp and F2 = +12pp as independent contributions. GLM2 shows these were an averaging artefact: the signal is concentrated in the F1×F2 conjunction, and both main effects are near-zero in isolation.

8. Analysis files

File	Purpose
`Tools/Analysis/decoupled_dots_glm2.py`	Fits the model; produces forest plot PDF
`Agents/Figures/decoupled_dots_glm2.pdf`	Forest plot: log-odds (left) + AME in pp (right)
`Agents/Literature/decoupled_dots_results.md` §3.6	Results table and narrative
`Agents/WriteUps/decoupled_dots_results.pdf`	Full rendered write-up